

Patient Independent MI Localization Project Summary

Implementation and evaluation of vectorcardiographic tensor decomposition and machine learning

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Guide Dr Shivnarayan Patidar | Status 1 October 2026 | Repository github.com/Kumaryan12/major_project

Executive summary

We implemented an end-to-end reproduction of Zhang et al. for myocardial infarction location classification from vectorcardiographic signals. We then replaced random heartbeat splitting with patient-disjoint evaluation, compared classifiers and feature representations, tested the selected model on PTB-XL, and investigated measured versus ECG-derived VCG. The project now has a reproducible research baseline and a clear improvement path, but it has not yet produced a clinically reliable or state-of-the-art model.

The central result is that evaluation design changes the conclusion. The reproduced tensor pipeline reaches 99.14% accuracy when beats from the same patients can appear in training and testing, but only 41.39% when test patients are completely unseen.

99.14% Random beat accuracy	41.39% Unseen patient split beat accuracy	0.487 Best PTB patient macro F1	0.242 External PTB XL patient macro F1
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Work completed

- Reproduced the paper cohort and preprocessing on the PTB Diagnostic ECG Database.
- Implemented wavelet denoising, R peak detection, beat extraction, VCG tensors, Tucker features and a 500 tree random forest.
- Added leakage-free patient grouped evaluation and a six-class patient-level benchmark.
- Compared random forest, linear SVM and XGBoost with nested patient-disjoint tuning.
- Evaluated 12 signal and feature representations and confirmed the strongest PTB candidate.
- Completed locked external testing on PTB-XL and a paired Frank versus Kors VCG study.
- Added automated tests, fixed experiment configurations, machine-readable outputs, documentation, a professor report and a presentation.

Current status

The research pipeline and all experiments listed above are complete and pushed to GitHub through commit **34eea1c**. The deep-learning and tensor-plus-learned-feature improvement phase remains future work.

Implemented method and reproducibility

The implementation follows the published signal-processing and tensor-decomposition design while recording choices that the paper did not fully specify.

Stage	Implemented method	Output
Data	PTB measured Frank X Y Z signals at 1000 Hz	178 patients and 425 records
Preprocessing	bior6.8 denoising and Pan Tompkins style R peak detection	60507 complete beats
Beat tensor	651 samples by 3 leads by 8 original and wavelet slices	3 x 651 x 8 tensor
Paper feature	Tucker compression of the time mode to rank 3	72 features per beat
Paper classifier	500 tree bootstrap random forest	Beat and patient grouped evaluation
Improved feature	Eight statistics for every lead and wavelet slice	192 features per beat

Reproduction quality

The final cohort contains 126 MI patients and 52 healthy controls. We detected 60507 beats, while the paper reports 60527, a difference of 20 beats or 0.033%. The paper does not publish its code, exact R peak annotations, wavelet threshold rule, fold assignments or all TreeBagger options, so exact numerical identity is not possible.

The implementation is deterministic and configuration-driven. Patient folds, experiment settings, metrics, predictions and confusion matrices are stored in the repository. Raw datasets, feature caches and fitted models are excluded because of size.

Evaluation protocols

Protocol	Purpose	Patient overlap
Random beat 10 fold	Reproduce the paper-style result	Almost every patient appears on both sides
Patient grouped 10 fold	Measure generalization to unseen patients	Zero overlap
Nested six-class benchmark	Tune models without using outer test patients	Zero overlap in outer and inner splits
Locked PTB-XL fold 10	Test transportability without external tuning	External patients only

Why the six-class task was required

The original 12-class task includes MI locations represented by only one patient. A model cannot learn such a class when its only patient is held out. The primary patient benchmark therefore retains AMI, ALMI, ASMI, IMI, ILMI and healthy control, each with at least 10 patients. This gives 164 patients and 55863 beats. Patient-level macro F1 is the primary endpoint because it weights all six classes equally.

Experimental results

Experiment	Cohort and protocol	Accuracy	Macro F1
Paper-style tensor baseline	178 PTB patients random beat folds	99.14%	0.991
Same tensor baseline	178 PTB patients grouped by patient	41.39%	0.191
Tucker plus XGBoost	164 PTB patients nested six-class CV	46.95%	0.403
Wavelet statistics plus XGBoost	164 PTB patients nested six-class CV	54.88%	0.487
PTB to PTB-XL external test	1092 PTB-XL patients locked fold 10	56.23%	0.242

Patient-independent benchmark

On the same six-class PTB task, random forest obtained 0.380 patient macro F1, linear SVM 0.366 and XGBoost 0.403. Their uncertainty intervals overlap, so the current cohort does not prove that one classifier is definitively superior. Healthy controls are consistently easier than MI locations.

Feature study

Twelve representations were evaluated on frozen patient folds. Wavelet statistics without Tucker compression performed best. In the nested comparison, patient macro F1 increased from 0.403 to 0.487 and patient accuracy increased from 46.95% to 54.88%. The paired bootstrap interval for the macro F1 improvement was +0.003 to +0.165. Because the representation was selected after exploratory PTB comparisons, the improvement still needs independent confirmation.

External validation

The fixed PTB model was tested once on 1185 ECG records from 1092 PTB-XL patients. Twelve-lead ECG was converted to estimated VCG with the Kors regression before applying the same feature pipeline. Patient macro F1 fell to 0.242. The cohort is 72.0% healthy, and an always-healthy classifier would achieve 71.98% accuracy, higher than the model's 56.23%. This confirms that raw accuracy is misleading and that the current model does not transport reliably.

Measured and derived VCG analysis

Training domain	Test domain	Patient macro F1
Measured Frank VCG	Measured Frank VCG	0.469
Measured Frank VCG	Kors-derived VCG	0.422
Kors-derived VCG	Kors-derived VCG	0.467

The measured-to-derived reduction is 0.046 macro F1, with a 95% interval from -0.108 to +0.017. It may contribute to the external gap, but it does not explain the much lower PTB-XL result. Device, recording length, labels, prevalence and case mix also change.

Conclusions and next research phase

What the project establishes

- The published pipeline has been reproduced closely under its beat-level evaluation setup.
- Random beat splitting substantially overestimates performance for the intended unseen-patient task.
- Wavelet statistics improve the current PTB patient-level result, but the choice is not independently confirmed.
- The PTB-trained model performs poorly on a locked PTB-XL cohort and is not suitable for clinical use.
- The Kors conversion explains only part of the domain difference within the limits of the paired PTB analysis.

Limitations

Limitation	Effect on interpretation
Small MI classes	Patient-level estimates remain uncertain, especially for rare locations.
PTB-guided feature selection	The wavelet-statistics gain requires a new untouched confirmation set.
Different databases	PTB and PTB-XL differ in signals, labels, duration, devices and prevalence.
Sparse external labels	The locked test includes only five ALMI patients.
Research prototype	No diagnostic, deployment or state-of-the-art claim is justified.

Recommended next milestone

Freeze the six-class label definition, patient folds and patient macro F1 endpoint with the project guide. Then implement a compact 1D CNN or ResNet on VCG beats and a hybrid model that combines learned signal features with tensor or wavelet features. Compare all models on exactly the same patient folds, report paired uncertainty, and reserve a genuinely untouched dataset for confirmation.

Decisions needed from the project guide

- Confirm whether the final project will retain the six-class endpoint or use a different clinical label grouping.
- Approve the deep-learning baseline, hybrid design and available compute budget.
- Agree on the independent validation plan before further model selection.

Repository and supporting material

Repository: github.com/Kumaryan12/major_project. The repository contains setup instructions, reproducible commands, fixed configurations, results, tests, the detailed progress report and the presentation deck. Latest included commit: 34eea1c.

Reference

Zhang et al. Automated Localization of Myocardial Infarction From Vectorcardiographic via Tensor Decomposition. IEEE Transactions on Biomedical Engineering. 2023;70(3):812-823. DOI 10.1109/TBME.2022.3202962.